

Advanced Undergraduate Laboratory Courses

AY2025-2026

Chem 100r Experimental Chemistry and Chemical Biology, Spring 2026, T&Th 12-5pm

This course is hands-on and project-based, so you'll be working in small groups on ongoing research projects in synthetic organic chemistry following recently published peer-review papers. The idea is to tackle the kinds of experimental problems chemists face when they're working with potentially therapeutic organic compounds. You'll get experience with synthesizing, isolating, purifying, characterizing, and identifying these compounds.

As you conduct research, you'll build both your technical lab skills and your understanding of the theory behind multistep organic synthesis and various spectroscopy techniques including multinuclear Nuclear Magnetic Resonance (NMR) spectroscopy, Gas Chromatography (GC), High Performance Liquid Chromatography (HPLC), and High-Resolution Mass Spectrometry (HRMS). You'll also work on problem sets related to spectroscopy, which will help you figure out how to analyze the data you collect and the progress you make on the research problems.

To keep things organized, you'll use electronic lab notebooks to document your experiments and findings. Additionally, you'll present your results at group meetings and put together a research paper as if you were submitting it for peer reviewed publication. Plus, you'll have the opportunity to practice communicating your science with your classmates and helping you to learn to discuss scientific problems just like you would in an academic research lab.

So, if you're excited about actually *doing* research, working closely with others, and building both your experimental and science dissemination skills, this course will give you a great taste of what real scientific research looks like.

Course Prerequisites: Chem 20 and 30, Chem 17 and 27, or Chem S-20ab.

Chem 135 Experimental Organic Chemistry, Spring 2026, M&W 12-5pm

If you're looking to really dive into the world of multi-step organic synthesis, organometallic synthesis, and catalysis Chem 135 is a great choice. In this course, you'll learn all about synthesis, isolation, purification, characterization, and identification of air, and light sensitive organic and organometallic compounds. It's a hands-on lab course so you'll develop synthetic laboratory skills including glovebox and Schlenk line while also understanding the theory and practice behind organotransition metal chemistry and spectroscopy.

Are you curious about reactions that are well known in medicinal chemistry, like palladium-catalyzed cross-coupling reactions? In Chem 135, we cover Suzuki-Miyaura, Sonogashira, and Buchwald-Hartwig reactions in detail. Plus, if you've ever wanted to try improving reactions from published research like incorporating Microwave Irradiation instead of just using traditional heating, you'll get that opportunity here. For instance, we do an in-depth exploration of the recently developed Aminative Suzuki-Miyaura reaction using Microwave Synthesis, which lets you explore beyond what's already reported.

You'll also get hands-on experience with advanced analytical techniques such as multinuclear Nuclear Magnetic Resonance (NMR) spectroscopy, Gas Chromatography (GC), High Performance Liquid Chromatography (HPLC), and High-Resolution Mass Spectrometry (HRMS). So, if you're interested in building your lab skills and understanding how organic synthesis works both in theory and practice, Chem 135 would be a great fit for you.

Course Prerequisites: Chem 20 and 30, Chem 17 and 27, or Chem S-20ab.

Chem 145 Experimental Inorganic Chemistry, Fall 2025, T&Th 12-5pm

If you're interested in getting hands-on experience with inorganic chemistry in the lab, Chem 145 could be what you're looking for. In this course, you'll learn all about synthesizing, purifying, and characterizing inorganic compounds. Additionally, you'll practice proper and safe ways to handle and store compounds that are sensitive to air, light, or water including how to use an inert-atmosphere glovebox and a Schlenk line. Chem 145 is also a great way to get comfortable with equipment and techniques used in academic research labs. In Chem 145, you won't just make these compounds, you'll also analyze them using a whole range of tools and methods, like Infrared Spectroscopy (IR) and Electron Paramagnetic Resonance spectroscopy (EPR), Nuclear Magnetic Resonance (NMR) Spectroscopy, Mössbauer Spectroscopy, electronic absorption spectroscopies, and electrochemical techniques. So, Chem 145 is a mix of hands-on synthesis and modern characterization techniques.

Alongside the lab work, you'll also get in-depth practice with storing your research properly. You'll keep clear, comprehensive entries in an electronic lab notebook (hosted at Harvard Wiki), making sure your data and observations are well-documented. This is what researchers do in an academic research group. Additionally, you'll build your skills in reading and understanding of primary scientific literature, writing your own research papers and then, as a whole class submitting that paper to peer-reviewed journals. This experience will teach you how to write an abstract, introduction, research and development, and conclusion of a scientific paper and get experience on manuscript submission and review process. Also, you will have the opportunity to prepare in depth Supporting Information of your paper. Finally, you will get hands on experience in learning how to search for information using tools like SciFinder and Web of Science.

So, if you're hoping to develop both your practical inorganic lab techniques and your ability to communicate your science, Chem 145 is a great option.

Course Prerequisites: Chem 20, Chem 40.

Chem 165 Experimental Physical Chemistry, Spring 2026, M&W 12-5pm

If you're someone who's curious about how experiments in physical chemistry are actually done and you enjoy working with cool scientific equipment, this course could be just what you're looking for. The main goal here is to give you a solid understanding of both classic and cutting-edge methods in experimental physical chemistry. That means you'll not only learn the techniques,

but also get hands-on experience with experiments that are helping scientists answer some of today's big questions.

In this class, you'll get to work with tools and methods that are widely used in laboratories for many disciplines of science, not just chemistry. That can be really fun, especially if you like the challenge of getting experimental setups to work and discovering firsthand how they reveal the secrets of complex molecular systems. Many of the techniques you'll be introduced to are actually the foundation of modern experimental nanoscience, so the skills you pick up here can be applied in many different research areas.

You'll also get a practical introduction to a wide range of physical methods used in chemistry and chemical physics labs. To tie it all together, you'll use computers for data collection and analysis skills that researchers rely on every day. Additionally, you will be assigned an independent research project that you can design from beginning to end. You will also have the opportunity to present your project at a poster session as a part of the Undergraduate Research Symposium organized by the Chemistry and Chemical Biology Department. In this module, you will have the opportunity to incorporate multinuclear Nuclear Magnetic Resonance (NMR) Spectroscopy, Infrared Spectroscopy (IR), High Performance Liquid Chromatography (HPLC), Electrochemistry, Ion Sensing Electrodes, 3D printing into your studies.

So, if you're excited about learning how different experiments work and you want practical experience using key techniques that are central to modern science, this class could be a great fit for you.

Course Prerequisites: Physical Sciences 11 or equivalent preparation in chemical bonding and fundamental principles; two introductory physics courses (e.g., Physical Sciences 2 and Physical Sciences 3). Chem 160 is recommended but not strictly required.

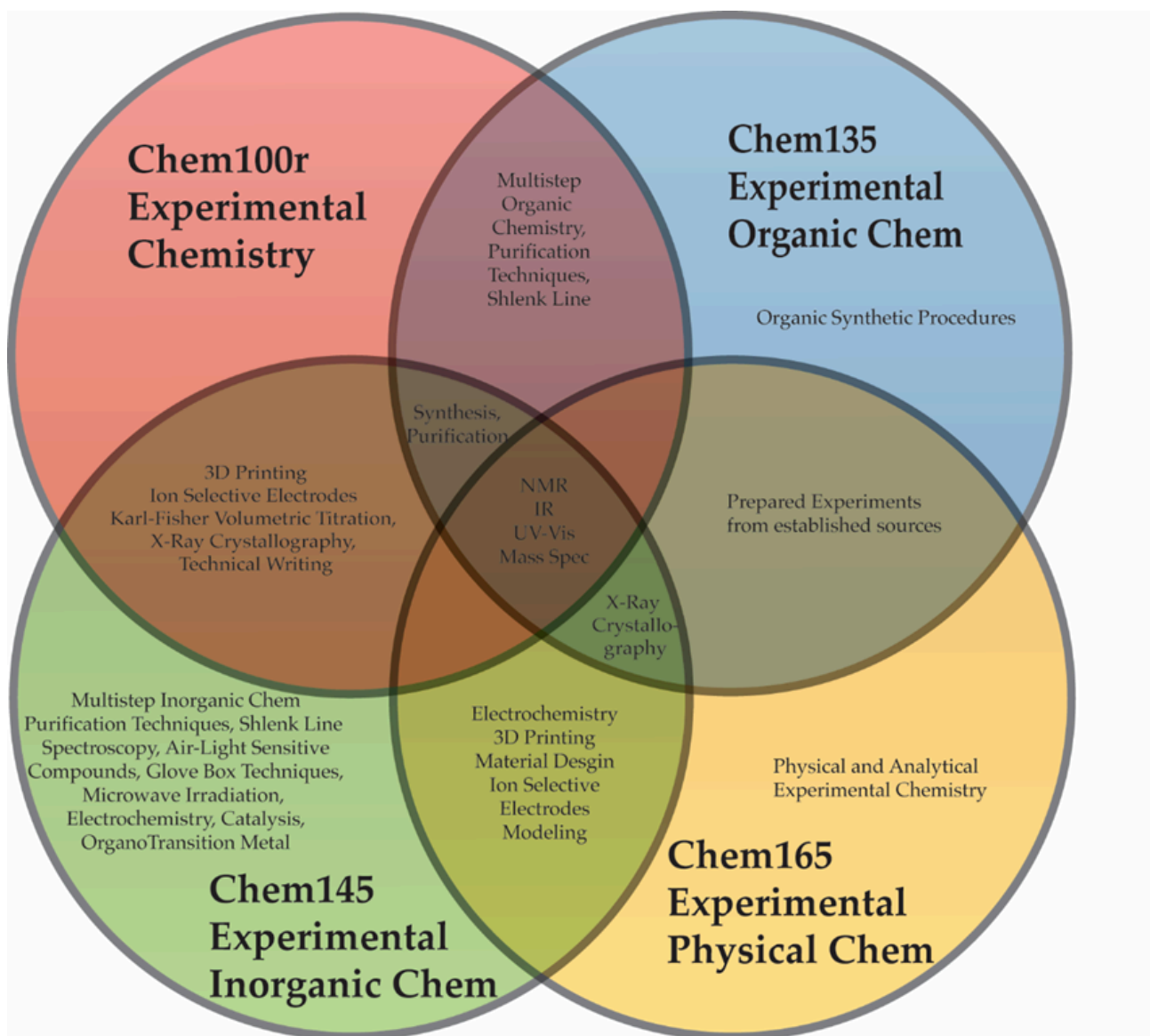


Figure 1. FAS Chem & Chem Biology Advanced Undergraduate Lab Courses.